Revision — 2.1.4 Thinking Logically

OCR H446 — one-page revision summary

Must know

- **Thinking logically** = identifying the **decision points** in a problem where the **flow of control** branches on a condition.
- A **decision point** is any place where execution can take different routes: `IF` / `ELSE`, `CASE` / `SELECT`, and loop tests (`WHILE` / `FOR` / `REPEAT UNTIL`).
- **Flow of control** = the order in which statements execute, shaped by these decision points.
- Each decision must be expressed as an **exact Boolean condition** (e.g. `IF seatsRequested <= seatsAvailable`), not a vague phrase like "if enough seats".
- Always state the outcome of **both branches** — what happens when the condition is **true** and when it is **false**.
- Conditions can combine with **AND / OR / NOT** for more complex logic (e.g. `IF age >= 18 AND balance > 0`).
- Getting the logic right (correct operators, boundaries, both branches) prevents errors such as off-by-one and unreachable code.

Key terms

Term	Definition
Thinking logically	Identifying decision points where flow of control branches
Decision point	A place where execution branches on a condition
Boolean condition	An expression evaluating to true or false
Flow of control	The order in which statements execute
Selection	Choosing between branches using `IF` / `CASE`
Iteration	Repeating steps controlled by a condition (loop test)
Logical operator	`AND`, `OR`, `NOT` used to combine conditions

Worked example / how to — apply logical thinking to a cinema booking

Decision points and their exact conditions:

- **Seat check:** `IF seatsRequested <= seatsAvailable` → **true:** proceed to payment; **false:** show "not enough seats" and stop.
- **Payment:** `IF paymentSuccessful = True` → **true:** confirm booking and update seat map; **false:** cancel and release the held seats.
- **Discount:** `IF age < 16 OR isStudent = True` → **true:** apply concession price; **false:** charge full price.

State each condition precisely and give **both** the true and false outcome for every branch.

Exam tips

- **Apply, don't define** — give the scenario's **exact Boolean condition** using real variables/operators, not "if there are enough".
- Always describe the outcome of **both branches** (true *and* false).
- Watch **boundary operators** (`<` vs `<=`) — the wrong one is a common logic error.
- Use **AND/OR/NOT** correctly when combining conditions; check the logic isn't unreachable or always-true.